

MUKUL SINGH

 My-portfolio

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Education

Dr. APJ Abdul Kalam Technical University

Bachelor of Technology in Computer Science C.G.P.A - 7.5

Aug 2021 – June 2024

Lucknow, Uttar Pradesh

Experience

EitBiz Technologies

July 2024 – Present

Backend-Focused Full Stack Developer

- Architected and developed scalable RESTful APIs using Node.js and Express.js for production applications including the Munch Market app backend, handling real-time data flow across mobile and web clients.
- Modeled and tuned database schemas across PostgreSQL and MongoDB, leveraging PostGIS geospatial queries, compound indexes, and aggregation pipelines to support location-based search and complex data retrieval.
- Configured cloud infrastructure including EC2 instances, S3 buckets, and load balancers on AWS; utilized GCP services for serverless functions and managed database hosting.
- Engineered backend infrastructure by integrating Redis for caching, session management, and rate limiting, containerizing services with Docker, and implementing authentication and authorization using JWT, OAuth 2.0, and Firebase Auth.

Projects

 **Zemora—Ride & Rescue Platform** | *Node, Express, Firebase, Redis, Socket.IO, PostgreSQL* **February 2026**

- **Real-Time Location & Communication:** Engineered a driver tracking system using Redis to cache live coordinates with batched PostgreSQL persistence every 30 seconds, cutting database write load significantly; integrated Socket.IO for in-app chat and ride events, plus Firebase Cloud Messaging for push notifications.
- **Dual-Flow Ride Engine:** Composed a unified backend supporting two flows — standard rides (accept → pickup passenger → drop at destination) and vehicle rescue (reach stranded car → tow → deliver to rescue point) — plus an SOS alert feature that routes emergencies directly to the admin panel for immediate action.
- **Live Ride Sharing:** Built a ride-sharing feature where riders share a generated URL with family/friends, opening a live tracking page that displays the driver's location, pickup, and destination on an interactive map.

 **Munch Market — Food Marketplace Backend** | *Node.js, Express.js, PostgreSQL, Socket.IO, Firebase* **October 2025**

- **Dual-Role Platform:** Architected the complete backend for an Airbnb-style food marketplace, designing a unified account model that lets users switch between buyer and home-chef roles within a single session, with role-based permissions and separate order/listing workflows under one identity.
- **Geospatial Search:** Implemented PostGIS-powered geospatial queries with GIST indexing to drive the “kitchens near you” feature, returning nearby home chefs ranked by distance with estimated delivery times.
- **Database Design:** Engineered normalized PostgreSQL schemas across 12 tables — user profiles, kitchen listings, menu items, orders, reviews, and bookings — with composite and partial indexing on high-read query paths (search, listings, order history) to maintain performance under load.
- **Real-Time Communication:** Engineered a Socket.IO-based chat system between buyers and chefs with instant messaging, read receipts, and typing indicators; integrated Firebase Cloud Messaging for cross-platform push notifications and SendGrid for transactional emails covering order confirmations, account verification, and password resets.

 **Complex.so** | *React.js, Node.js, PostgreSQL, Docker*

November 2024

- **Real-Time Chats:** Established a robust real-time chat system using Node and Socket, featuring instant messaging, typing indicators, read receipts, and message timestamps, ensuring seamless and efficient user communication.
- **Intuitive User Interface:** Built a responsive React.js interface with reusable component architecture, supporting core user flows for across desktop and mobile viewports.
- **Store Management:** Implemented a modular state management layer using Redux Toolkit in a React.js application, with centralized slices for reactive data, selectors, and async thunks to streamline data flow across components.

Technical Skills

Languages: JavaScript, TypeScript, C, Java, SQL

Backend & Frontend: Node.js, Express.js, React.js, Socket.IO, REST APIs, JWT, OAuth 2, HTML, Next.js, Tailwind CSS

Databases & Caching: PostgreSQL, MongoDB, Redis, Firebase

Cloud, DevOps & Tools: AWS (EC2, S3), GCP, Docker, Git, Nginx, SendGrid, Mapbox, Postman

Certifications & Extracurricular

- Completed certifications in Web Development and Java (Apna College), and API Testing (Postman).
- Secured AIR 2707 in CodeKaze Hiring Challenge by Coding Ninjas, competing against thousands of developers nationwide.
- Solved 100+ problems on LeetCode, demonstrating proficiency in algorithms, data structures, and analytical reasoning.